

Day 6 — Indian Mathematics

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Students name 3 specific Indian mathematical contributions with dates and articulate why each matters.

Materials

- Whiteboard
- Printed extract: 1 verse from Bhāskara II's *Līlāvatī* (12th c. CE) in translation

Vocabulary

- *gaṇita* — “calculation, mathematics”
- *śūnya* — “zero, void”
- *aṅka* — “digit”

Flow

Warm-up (5 min)

Daily prompt + recap: 2 students share one math fact they remember about India from prior knowledge.

Core (25 min)

1. Three specific contributions to remember:

(a) Zero (śūnya) as a number — Bakhshali manuscript (~7th c. CE — older fragments possibly earlier). Distinct from “zero as placeholder” (which existed earlier in Babylonian and Maya systems). The Indian innovation: zero participates in arithmetic operations (you can add, subtract, multiply, divide... with zero).

(b) Decimal place-value system — by 6th c. CE the Indian numeral system (with positional values for ones, tens, hundreds, etc.) was in widespread use. Adopted by Arab mathematicians (Al-Khwārizmī, ~825 CE), reached Europe by ~12th c. via Fibonacci (1202 CE).

(c) Infinite series (centuries before Newton) — Mādhava of Saṅgamagrāma (~14th c. CE, Kerala school) computed infinite series for π , sine, cosine. Specifically:

$$\pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right)$$

(This is now called the “Leibniz series” but Mādhava had it 300 years earlier.)

2. **A named text — Bhāskara II's *Līlāvātī* (1150 CE).** A textbook on arithmetic and algebra written for his daughter (the name “*Līlāvātī*” is reportedly hers). Includes problems on geometry, sequences, equations, mensuration. Read one short problem aloud (in translation).
3. **What modern math doesn't credit Indian sources for (but should):**
 - The number zero as a number (not just placeholder).
 - The “Leibniz” series for π (Mādhava had it).
 - Several trigonometric identities.
4. **What Indian math DIDN'T do:**
 - Did not develop full coordinate geometry (that's Descartes, 17th c. European).
 - Did not develop calculus formally (though Mādhava came very close with the series).
 - Did not develop modern algebraic notation.

Both lists matter for honesty.

Activity (10 min) — Compute the Mādhava series

Class together. On the board: - Start at 1. - Subtract $1/3$. Add $1/5$. Subtract $1/7$. Add $1/9$ - After 4 terms: $1 - 0.333 + 0.2 - 0.143 = 0.724$ - Multiply by 4: 2.896 (close to $\pi = 3.14$) - After 10 terms it gets closer.

The series converges SLOWLY — that's one reason it wasn't useful as a calculation method. But proving the *form* was a huge mathematical achievement.

Wrap-up (5 min)

- Preview Day 7: “*Tomorrow — Indian astronomy. Aryabhata.*”

Student-facing content

Three Indian Mathematical Contributions

1. **Zero as a number** — Bakhshali manuscript, ~7th c. CE.
2. **Decimal place-value system** — by 6th c. CE; adopted by Arab mathematicians (~825 CE); reached Europe (~12th c.).
3. **Infinite series for π** — Mādhava of Saṅgamagrāma, ~14th c. CE.

$$\pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right)$$

(Now sometimes called the “Leibniz series” — Mādhava had it ~300 years before Leibniz.)

A named text: **Bhāskara II's *Līlāvātī*** (1150 CE).

Homework

- Compute 10 terms of the Mādhava series for π . How close do you get?

Differentiation

- **Tier up:** Read about the Madhava-Leibniz attribution debate. Why is the European name still standard?
- **Tier down:** Just remember the three contributions + their approximate dates.

Teacher notes

- The Leibniz-Mādhava attribution is a recurring honest-credit issue. Mainstream math history is increasingly correcting this; some textbooks now use “Madhava-Leibniz series.”
- Bhāskara II’s *Līlāvātī* problems are charming — many are presented as little stories. If time permits, read one or two more.

Citation(s) used

- Plofker (2009), Chapters 4–7.
- Bhāskara II, *Līlāvātī* (1150 CE) — translation by Colebrooke (1817) or Patwardhan, Nainpally, Singh (2001).
- Hayashi (1995) on the Bakhshali manuscript.